

Curriculum Vitae

Ting S. Li

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Personal Information

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Academic Qualifications

2010 – 2016	Ph.D., Physics, Texas A&M University, USA <i>Thesis:</i> Exploring Milky Way Halo Substructures with Large-area Sky Surveys <i>Advisors:</i> Dr. Darren L. DePoy; Dr. Jennifer L. Marshall (Co-chair)
2008 – 2010	M.S., Space Science and Technology (SpaceMaster – Erasmus Mundus Course) Luleå University of Technology, Kiruna, Sweden Université Paul Sabatier Toulouse III, Toulouse, France <i>Thesis:</i> Design of the High Energy Particle Instrument for Electrons for the Energization and Radiation in Geospace Mission <i>Advisor:</i> Masafumi Hirahara, University of Tokyo
2004 – 2008	B.S., Physics, Fudan University, Shanghai, China <i>Thesis:</i> Study of Quasar Accretion Disk with Microlensing; <i>Advisor:</i> Feng Yuan Minor: Diplomacy, Fudan University (2004–2008)

Employment

2021 – present	Assistant Professor, Department of Astronomy and Astrophysics, University of Toronto, Toronto, Canada
2019 – 2021	NASA Hubble Fellowship Program Einstein Fellow & Carnegie–Princeton Fellow, Carnegie Observatories (Pasadena, CA) and Department of Astrophysical Sciences, Princeton University (Princeton, NJ)
2016 – 2019	Leon M. Lederman Fellow in Experimental Physics, Fermi National Accelerator Laboratory, Batavia, IL, USA
2011 – 2016	Research Associate, Department of Physics & Astronomy, Texas A&M University, College Station, TX, USA

Research Interests

Near-field Cosmology; Galactic Archaeology; Dark Matter; Dwarf Galaxies; Stellar Streams; Metal-poor Stars; Stellar Populations; Ground-based Instrumentation; Multi-object Spectroscopy; Modern Sky Surveys; Big Data and Survey Science.

Research Grants

Science Projects

- **Canadian Space Agency JWST Cycle 5:** *The Faint Frontier: Pushing Proper Motion Limits with JWST+Gaia* (JWST Legacy AR program 12274; program PI: K. McKinnon, postdoc mentee). Role — CSA PI. (2026 – 2029, \$119,163 CAD; JWST science program approved, CSA funding decision pending)
- **Canadian Space Agency JWST Cycle 5:** *Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars* (JWST GO program 10246; program PI: N. Sandford, postdoc mentee). Role — CSA PI. (2026 – 2028, \$98,890 CAD; JWST science program approved, CSA funding decision pending)
- **Fields Institute, Dunlap Institute, and CITA:** Workshop funding for *Near-Field Cosmology in the Era of Big Data: Local Group and Beyond*, Toronto, July 2026. Role — Lead organizer. (2026, \$17,000 CAD total: \$10,000 Fields Institute + \$5,000 Dunlap Institute + \$2,000 CITA)
- **NSERC Alliance International Collaboration Grant:** *Probing Dark Matter with Astrophysical Observations*. Role — PI. (2026 – 2029, \$282,000 CAD)
- **Research Corporation for Science Advancement (Scialog, Early Science with the LSST):** *Measuring Metallicities of RRLs in the Milky Way Vicinity*. Role — Co-PI. (2026 – 2027, \$60,000 USD to U of T; \$120,000 USD total with co-PI N. Hernitschek)
- **Canadian Space Agency JWST Cycle 3:** *A Closer Look at the Formation and Evolution of M31's Inner Disk* (JWST GO program 4735; program PI: N. Sandford, postdoc mentee). Role — CSA PI. (2024 – 2026, \$97,900 CAD)
- **NSERC Alliance International Catalyst Grant:** *Towards a Unified Framework for Dwarf Galaxy Kinematics*. Role — PI. (2023 – 2024, \$25,000 CAD)
- **U. of Toronto Data Science Institute Catalyst Grant:** *Spectroscopy by the Millions: A Fast, Reproducible Framework to Yield Chemical Compositions of 4 Million Stars*. Role — Co-PI. (2023 – 2025, \$100,000 CAD)
- **NSERC Discovery Grant:** *Near Field Cosmology with Milky Way's Satellite Galaxies and Stellar Streams*. Role — PI. (2022 – 2027, \$155,000 CAD)
- **U. of Toronto Connaught New Researcher Award:** *Probe the Nature of Dark Matter with Milky Way's Satellite Galaxies*. Role — PI. (2022 – 2024, \$20,000 CAD)
- **Australian Research Council Discovery Project:** *Seeing Dark with Light: Revealing the Milky Way with Stellar Streams*. Role — Co-I. (2022 – 2026, \$700,000 AUD total award, held at Australian institutions)
- **NASA Hubble Fellowship Program:** *Constraining Dark Matter with Stellar Streams and Dwarf Galaxies*. Role — Science PI. (2019 – 2021, \$340,000 USD)

Instrumentation Projects

- **Canada Foundation for Innovation (CFI) + Ontario Research Fund (ORF):** *A Canadian Contribution to the ANDES instrument for the European Extremely Large Telescope*. Role — PI of the ORF grant (\$3.8M CAD); Co-I of the CFI grant (\$11.3M CAD; co-PIs R. Doyon, Université de Montréal, and A. Man, UBC). Total project: \$22.6M CAD (CFI + matching provincial funds). (2026 –)
- **Dunlap Seed Fund:** *DMD-MOS Development for Astronomical Observation*. Role — Admin-PI. (2024 – 2026, \$30,000 CAD)
- **U. of Toronto XSeed Joint Seed Grant:** *Constructing the Human Olfactory System ex vivo*. Role — Co-PI. (2023 – 2025, \$120,000 CAD)
- **Dunlap Seed Fund:** *Investigating Scientific CMOS Detectors for Astronomical Application*. Role — Co-PI. (2023 – 2025, \$75,000 CAD)
- **CFI/JELF (Canada Foundation for Innovation, John R. Evans Leaders Fund) + ORF (Ontario Research Fund):** *An Advanced Optical Instrumentation Laboratory for the Development of the Next Generation Spectroscopic Surveys*. Role — PI. (2022 – 2026, \$140,000 CFI + \$140,000 ORF)

Observation, Data Reduction, and Instrumentation Experience

Survey and Instrumentation Contributions

- **Dark Energy Survey (DES):** contributions to commissioning, operation, photometric calibration, and data releases; designed and built the Atmospheric Transmission Monitoring Camera (aTmCam), including 100+ nights of prototype testing and instrument commissioning. Earned DES Builder status (authorship and personal data rights) in 2015.
- **Dark Energy Spectroscopic Instrument (DESI):** contributions to the active optics (focus and alignment) system, commissioning, and science verification; target selection and survey planning for the DESI Milky Way Survey. Earned DESI external participant status in 2019 and Builder status in 2023.
- **HETDEX:** assembled over 80 instrument collimators of the Visible Integral-Field Replicable Unit Spectrographs (VIRUS) for the Hobby–Eberly Telescope Dark Energy Experiment.

Observing Experience

- Over 300 nights of hands-on observing experience on optical telescopes worldwide.
- Optical imaging: Blanco (4 m)/DECam (80+ nights).
- Optical spectroscopy: Magellan (6.5 m)/IMACS (60+ nights), Magellan/MIKE (20+ nights), Magellan/M2FS (8 nights); McDonald Observatory 2.1 m/ES2 (30+ nights) and 2.7 m/VIRUS-P (9 nights).
- Served as PI or co-PI for over 100 nights on 4–8 m class telescopes, including 100+ nights on the Anglo-Australian Telescope, 40+ nights on Magellan, and 100+ hours on the VLT.
- Served as admin PI of three JWST proposals (Cycle 3 and Cycle 5), with postdoc mentees as science PIs.

Honours and Awards

2024	OCPA Outstanding Young Researcher Award (OYRA Ardentec Prize) , International Organization of Chinese Physicists and Astronomers
2024	Scialog Fellow: Early Science with the LSST, Research Corporation for Science Advancement
2023	Dorothy Shoichet Women Faculty in Science Award of Excellence
2022	Rising Stars in Physics, Princeton University
2019 – 2021	NASA Hubble Fellowship
2019 – 2021	Carnegie–Princeton Fellowship
2016 – 2019	KICP Associate Fellow, University of Chicago
2016 – 2019	Leon M. Lederman Fellow in Experimental Physics
2015 – 2016	Mitchell Institute Graduate Fellowship
2013 – 2014	Dr. Chia-Lai Wang Memorial Scholarship, Texas A&M University
2008 – 2010	Erasmus Mundus Scholarship
2008	Graduated as First-Class Student, Fudan University
2004 – 2008	People’s Scholarship, Fudan University

Scholarly Output

Refereed

Total: 250+ refereed publications (2014 – 2026), including 11 first-author, 46 second/third-author, and 58 papers as 4th – 8th author with significant contributions.

h-index: 98

citations: 40,000+

For a complete publication list, see [ADS Library](#) or [Google Scholar](#).

Papers written under my direct supervision (by students and postdocs in my group, or by lead authors whose work I supervised within collaborations) are marked with underline.

First Author

11. **Li, T. S.**, Erkal, D., Pace, A. B., Yang, J., et al. “Boötes III is a Tidally Disrupting Ultra-Faint Dwarf Galaxy on an Eccentric Polar Orbit”, 2026, submitted to OJAp, [arXiv:2607.07803](#).
10. **Li, T. S.**, Ji, A. P., Pace, A. B., et al. “ S^5 : The Orbital and Chemical Properties of One Dozen Stellar Streams”, 2022, ApJ, 928, 30. [\[link\]](#)
9. **Li, T. S.**, Koposov, S. E., Erkal, D., et al. “Broken into Pieces: ATLAS and Aliqa Uma as One Single Stream”, 2021, ApJ, 911, 149. [\[link\]](#)
8. **Li, T. S.**, Koposov, S. E., Zucker, D. B., et al. “The Southern Stellar Stream Spectroscopic Survey (S^5): Overview, Target Selection, Data Reduction, Validation, and Early Science”, 2019, MNRAS, 490, 3508. [\[link\]](#)
7. **Li, T. S.**, Simon, J. D., Kuehn, K., et al. “The First Tidally Disrupted Ultra-Faint Dwarf Galaxy? — Spectroscopic Analysis of the Tucana III Stream”, 2018, ApJ, 866, 22. [\[link\]](#)
6. **Li, T. S.**, Simon, J. D., Pace, A. B., et al. “Ships Passing in the Night: Spectroscopic Analysis of Two Ultra-faint Satellites in the Constellation Carina”, 2018, ApJ, 857, 145. [\[link\]](#)
5. **Li, T. S.**, Sheffield, A. A., Johnston, K. V., Marshall, J. L., Majewski, S. R., et al. “Exploring Halo Substructure with Giant Stars”, 2017, ApJ, 844, 74. [\[link\]](#)
4. **Li, T. S.**, Simon, J. D., Drlica-Wagner, A., Bechtol, K., et al. “Farthest Neighbor: The Distant Milky Way Satellite Eridanus II”, 2017, ApJ, 838, 8. [\[link\]](#)
3. **Li, T. S.**, DePoy, D. L., Marshall, J. L., Tucker, D. L., Bernstein, G. M., et al. “Assessment of Systematic Chromatic Errors that Impact Sub-1% Photometric Precision in Large-Area Sky Surveys”, 2016, AJ, 151, 157. [\[link\]](#)
2. **Li, T. S.**, Balbinot, E., Mondrik, N., et al. “Discovery of a Stellar Overdensity in Eridanus–Phoenix in the Dark Energy Survey”, 2016, ApJ, 817, 135. [\[link\]](#)
1. **Li, T. S.**, Marshall, J. L., Lépine, S., Williams, P., Chavez, J. “Optical BVRI Photometry of Common Proper Motion F/G/K+M Wide Separation Binaries”, 2014, AJ, 148, 60. [\[link\]](#)

Second or Third Author (mostly led by mentees)

46. Nguyen, T., Necib, L., **Li, T. S.**, et al. “Dark Matter in Draco and Boötes I: Hints of a Core in an Ultra-Faint Dwarf from Simulation-Based Inference”, 2026, [arXiv:2606.26218](#).
45. Carlberg, R. G., **Li, T. S.**, Jarvis, E., Mohammed, N., et al. “Joint Modeling of GD-1 and C-19 as Old Streams”, 2026, submitted, [arXiv:2606.15863](#).
44. Byström, A., Koposov, S. E., **Li, T. S.**, et al. “Constraining the Galactic bar using the M92 stellar stream”, 2026, submitted, [arXiv:2605.07918](#).
43. Jarvis, E., **Li, T. S.**, Koposov, S. E., et al. “Characterizing the GD-1 Stream with DESI DR2 Data: Thin Stream and Hot Cocoon”, 2026, submitted, [arXiv:2604.20958](#).
42. Mohammed, N., Tang, J. Y., **Li, T. S.**, et al. “The Kinematically Hot, Extremely Metal-Poor C-19 Stellar Stream in DESI DR2”, 2026, ApJ, 1005, 200. [\[link\]](#)
41. Yang, Y., Lewis, G. F., **Li, T. S.**, et al. “Epicyclic density variations in the Indus stellar stream”, 2026, MNRAS, 547, stag488. [\[link\]](#)
40. Cerny, W., **Li, T. S.**, Pace, A. B., et al. “A Chemodynamical Census of the Milky Way’s Ultra-Faint Compact Satellites. I. A First Population-Level Look at the Internal Kinematics and Metallicities of 19 Extremely-Low-Mass Halo Stellar Systems”, 2026, submitted, [arXiv:2602.17652](#).
39. Yang, H., Wang, W., **Li, T. S.**, et al. “The Shape-velocity Alignment of Satellites Forged by Tidal Locking and Dynamical Friction”, 2026, ApJ, 999, 158. [\[link\]](#)
38. Limberg, G., Ji, A. P., **Li, T. S.**, et al. “ S^5 : Tidal Disruption in Crater 2 and Formation of Diffuse Dwarf Galaxies in the Local Group”, 2025, submitted, [arXiv:2512.02177](#).
37. Ding, J., Rockosi, C., **Li, T. S.**, et al. “The Draco Dwarf Spheroidal Galaxy in the First Year of Dark Energy Spectroscopic Instrument Data”, 2025, ApJ, 994, 134. [\[link\]](#)
36. Sandford, N. R., **Li, T. S.**, Koposov, S. E., et al. “Chemodynamics of Boötes I with S^5 : Revised Velocity Gradient, Dark Matter Density, and Galactic Chemical Evolution Constraints”, 2026, ApJ, 998, 47. [\[link\]](#)
35. Medina, G. E., **Li, T. S.**, Eadie, G. M., et al. “The mass of the Milky Way from outer halo stars measured by DESI DR1”, 2025, submitted, [arXiv:2508.19351](#).
34. Heiger, M. E., Ji, A. P., **Li, T. S.**, et al. “Not-so-heavy metal(s): Chemical Abundances in the Ultra-faint Dwarf Galaxies Eridanus IV and Centaurus I”, 2026, ApJ, 1002, 182. [\[link\]](#)
33. Pace, A. B., **Li, T. S.**, Ji, A. P., et al. “Spectroscopic Analysis of Pictor II: a very low metallicity ultra-faint

- dwarf galaxy bound to the Large Magellanic Cloud”, 2025, OJAp, 8, 112. [\[link\]](#)
32. Koposov, S., **Li, T. S.**, Allende Prieto, C., et al. “DESI Data Release 1: Stellar Catalogue”, 2026, OJAp, 9, 55260. [\[link\]](#)
 31. Medina, G. E., **Li, T. S.**, Allende Prieto, C., et al. “The DESI Y1 RR Lyrae catalog II: The metallicity dependency of pulsational properties, the shape of the RR Lyrae instability strip, and metal-rich RR Lyrae”, 2025, ApJ, in press, [arXiv:2505.10614](#).
 30. Kim, B., Koposov, S. E., **Li, T. S.**, et al. “Nearby stellar substructures in the Galactic halo from DESI Milky Way Survey Year 1 Data Release”, 2025, MNRAS, 540, 264. [\[link\]](#)
 29. Medina, G. E., **Li, T. S.**, Koposov, S. E., et al. “The DESI Y1 RR Lyrae catalog I: Empirical modeling of the cyclic variation of spectroscopic properties and a chemodynamical analysis of the outer halo”, 2025, ApJ, in press, [arXiv:2504.02924](#).
 28. Ma, P. X., Rogers, K. K., **Li, T. S.**, et al. “Toward Characterizing Dark Matter Subhalo Perturbations in Stellar Streams with Graph Neural Networks”, 2025, ApJ, 987, 96. [\[link\]](#)
 27. Awad, P., **Li, T. S.**, Erkal, D., et al. “ S^5 : New insights from deep spectroscopic observations of the tidal tails of the globular clusters NGC 1261 and NGC 1904”, 2025, A&A, 693, A69. [\[link\]](#)
 26. Simon, J. D., **Li, T. S.**, et al. “Eridanus III and DELVE 1: Carbon-rich Primordial Star Clusters or the Smallest Dwarf Galaxies?”, 2024, ApJ, 976, 256. [\[link\]](#)
 25. Yu, F., **Li, T. S.**, Speagle, J. S., et al. “The Power of High Precision Broadband Photometry: Tracing the Milky Way Density Profile with Blue Horizontal Branch stars in the Dark Energy Survey”, 2024, ApJ, 975, 81. [\[link\]](#)
 24. Hansen, T. T., Simon, J. D., **Li, T. S.**, et al. “Chemical Diversity on Small Scales: Abundance Analysis of the Tucana V Ultrafaint Dwarf Galaxy”, 2024, ApJ, 968, 21. [\[link\]](#)
 23. Usman, S. A., Ji, A. P., **Li, T. S.**, et al. “Multiple Populations and a CH Star Found in the 300S Globular Cluster Stellar Stream”, 2024, MNRAS, 529, 2413. [\[link\]](#)
 22. Heiger, M. E., **Li, T. S.**, Pace, A. B., et al. “Reading Between the (Spectral) Lines: Magellan/IMACS spectroscopy of the Ultra-faint Dwarf Galaxies Eridanus IV and Centaurus I”, 2024, ApJ, 961, 234. [\[link\]](#)
 21. Cerny, W., Drlica-Wagner, A., **Li, T. S.**, et al. “DELVE 6: An Ancient, Ultra-faint Star Cluster on the Outskirts of the Magellanic Clouds”, 2023, ApJ, 953L, 21. [\[link\]](#)
 20. Koposov, S. E., Erkal, D., **Li, T. S.**, et al. “ S^5 : Probing the Milky Way and Magellanic Clouds potentials with the 6D map of the Orphan-Chenab stream”, 2023, MNRAS, 521, 4936. [\[link\]](#)
 19. Bruce, J., **Li, T. S.**, Pace, A. B., et al. “Spectroscopic analysis of Milky Way outer halo satellites: Aquarius II and Boötes II”, 2023, ApJ, 950, 167. [\[link\]](#)
 18. Hansen, T. T., Simon, J. D., **Li, T. S.**, et al. “Evidence for multiple nucleosynthetic processes from carbon-enhanced metal-poor stars in the Carina dwarf spheroidal galaxy”, 2023, A&A, 674A, 180. [\[link\]](#)
 17. Cerny, W., Simon, J. D., **Li, T. S.**, et al. “Pegasus IV: Discovery and Spectroscopic Confirmation of an Ultra-Faint Dwarf Galaxy in the Constellation Pegasus”, 2023, ApJ, 942, 111. [\[link\]](#)
 16. Pace, A. B., Erkal, D., **Li, T. S.** “Proper Motions, Orbits, and Tidal Influences of Milky Way Dwarf Spheroidal Galaxies”, 2022, ApJ, 940, 136. [\[link\]](#)
 15. Ji, A. P., Koposov, S. E., **Li, T. S.**, et al. “Kinematics of Antlia 2 and Crater 2 from the Southern Stellar Stream Spectroscopic Survey (S^5)”, 2021, ApJ, 921, 32. [\[link\]](#)
 14. Jenkins, S., **Li, T. S.**, Pace, A. B., et al. “VLT Spectroscopy of Ultra-Faint Dwarf Galaxies. 1. Boötes I, Leo IV, Leo V”, 2021, ApJ, 920, 92. [\[link\]](#)
 13. Ji, A. P., **Li, T. S.**, Hansen, T. T., et al. “The Southern Stellar Stream Spectroscopic Survey (S^5): Chemical Abundances of Seven Stellar Streams”, 2020, AJ, 160, 181. [\[link\]](#)
 12. Simon, J. D., **Li, T. S.**, Erkal, D., et al. “Birds of a Feather? Magellan/IMACS Spectroscopy of the Ultra-faint Satellites Grus II, Tucana IV, and Tucana V”, 2020, ApJ, 892, 137. [\[link\]](#)
 11. Wan, Z., Lewis, G. F., **Li, T. S.**, et al. “The tidal remnant of an unusually metal-poor globular cluster”, 2020, Nature, 583, 768. [\[link\]](#)
 10. Koposov, S. E., Boubert, D., **Li, T. S.**, et al. “Discovery of a nearby 1700 km/s star ejected from the Milky Way by Sgr A*”, 2020, MNRAS, 491, 2465. [\[link\]](#)
 9. Ji, A. P., **Li, T. S.**, Simon, J. D., et al. “Detailed Abundances in the Ultra-faint Magellanic Satellites Carina II and III”, 2020, ApJ, 889, 27. [\[link\]](#)
 8. Shipp, N., **Li, T. S.**, Pace, A. B., et al. “Proper Motions of Stellar Streams Discovered in the Dark Energy Survey”, 2019, ApJ, 885, 3. [\[link\]](#)
 7. Koposov, S. E., Belokurov, V., **Li, T. S.**, et al. “Piercing the Milky Way: an all-sky view of the Orphan

- Stream”, 2019, MNRAS, 485, 4726. [\[link\]](#)
6. Pace, A. B. & **Li, T. S.** “Proper motions of Milky Way Ultra-Faint satellites with Gaia DR2 $\times$ DES DR1”, 2019, ApJ, 875, 77. [\[link\]](#)
 5. Erkal, D., **Li, T. S.**, Koposov, S. E., Belokurov, V., et al. “Modelling the Tucana III stream — a close passage with the LMC”, 2018, MNRAS, 481, 3148. [\[link\]](#)
 4. Nagasawa, D. Q., Marshall, J. L., **Li, T. S.**, et al. “Chemical Abundance Analysis of Three α -Poor, Metal-Poor Stars in the Ultra-Faint Dwarf Galaxy Horologium I”, 2018, ApJ, 852, 99. [\[link\]](#)
 3. Simon, J. D., **Li, T. S.**, Drlica-Wagner, A., Bechtol, K., et al. “Nearest Neighbor: The Low-mass Milky Way Satellite Tucana III”, 2017, ApJ, 838, 11. [\[link\]](#)
 2. Balbinot, E., Yanny, B., **Li, T. S.**, et al. “The Phoenix stream: a cold stream in the Southern hemisphere”, 2016, ApJ, 820, 58. [\[link\]](#)
 1. Simon, J. D., Drlica-Wagner, A., **Li, T. S.**, et al. “Stellar Kinematics and Metallicities in the Ultra-faint Dwarf Galaxy Reticulum II”, 2015, ApJ, 808, 95. [\[link\]](#)
- 4th – 8th Author (with significant contributions)**
58. Heiger, M. E., Ji, A. P., Speagle, J. S., **Li, T. S.**, et al. “It’s a matter of time: Empirical Constraints on Supernova Yields and Delay Times from Dwarf Spheroidal Galaxies”, 2026, ApJ, in press, [arXiv:2602.22333](#).
 57. Lambert, M., Rockosi, C. M., Koposov, S. E., **Li, T. S.**, et al. “Signatures of a Tidally Induced Spiral Arm at the Anticenter of the Milky Way and a Kinematically Extended Anticenter Stream Using DESI DR2”, 2026, AJ, 171, 292. [\[link\]](#)
 56. Li, S., Wang, W., Koposov, S. E., Amarante, J. A. S., Deason, A. J., Valluri, M., **Li, T. S.**, et al. “The velocity field of our Milky Way outer stellar halo based on DESI DR2”, 2026, ApJ, 1005, 19. [\[link\]](#)
 55. Bissonette, D., Ji, A. P., Simon, J. D., Bland-Hawthorn, J., Chiti, A., Geha, M., **Li, T. S.**, Frebel, A., Luna, A. M. “The Metallicity Distribution of the Ultra-Faint Dwarf Galaxy Segue 1”, 2026, submitted, [arXiv:2605.12810](#).
 54. Usman, S. A., Ji, A. P., Rodriguez, J., Simpson, J. D., Martell, S. L., **Li, T. S.**, et al. “Chemical Abundances in the Metal-Poor Globular Cluster ESO 280-SC06: A Formerly Massive, Tidally Disrupted Globular Cluster”, 2026, OJAp, 9, 55387. [\[link\]](#)
 53. Webber, K. B., Hansen, T. T., Marshall, J. L., Ji, A. P., **Li, T. S.**, et al. “Detailed Chemical Abundance Analysis of the Brightest Stars in the Turranburra and Willka Yaku Stellar Streams”, 2026, ApJ, 998, 114. [\[link\]](#)
 52. Qiu, T., Wang, W., Koposov, S. E., **Li, T. S.**, et al. “The Binary Fraction of Stars in the Dwarf Galaxy Ursa Minor via Dark Energy Spectroscopic Instrument”, 2026, ApJ, 1000, 33. [\[link\]](#)
 51. Li, S., Wang, W., Koposov, S. E., Amarante, J. A. S., Deason, A. J., Sandford, N. R., **Li, T. S.**, et al. “The Milky Way Stellar Halo Is Twisted and Doubly Broken: Insights from DESI DR2 Milky Way Survey Observation”, 2026, ApJ, 999, 108. [\[link\]](#)
 50. Tan, C. Y., Drlica-Wagner, A., Pace, A. B., Cerny, W., Nadler, E. O., Doliva-Dolinsky, A., Anbajagane, D., **Li, T. S.**, et al. “DELVE Milky Way Satellite Galaxy Census. I. Satellite Population and Survey Selection Function in DES, DELVE, and Pan-STARRS”, 2026, ApJ, 1000, 87. [\[link\]](#)
 49. Cohen, B., Ji, A. P., Ferguson, P. S., Koposov, S. E., Drlica-Wagner, A., Li, A. P., **Li, T. S.**, et al. “Sifting for a Stream: The Morphology of the 300S Stellar Stream”, 2026, OJAp, 9, 61089. [\[link\]](#)
 48. Yang, H., Wang, W., Zhu, L., **Li, T. S.**, et al. “The Dark Matter Content of Milky Way Dwarf Spheroidal Galaxies: Draco, Sextans, and Ursa Minor”, 2025, ApJ, 993, 249. [\[link\]](#)
 47. Luna, A. M., Ji, A. P., Chiti, A., Simon, J. D., Kelson, D. D., Go, M., Limberg, G., **Li, T. S.**, et al. “A Bimodal Metallicity Distribution Function in the Ultra-Faint Dwarf Galaxy Reticulum II”, 2025, OJAp, 8, 47696. [\[link\]](#)
 46. Aganze, C., Chandra, V., Wechsler, R. H., **Li, T. S.**, et al. “The Cocytos Stream: A Disrupted Globular Cluster from our Last Major Merger?”, 2025, [arXiv:2504.11687](#).
 45. Yang, Y., Lewis, G. F., Erkal, D., **Li, T. S.**, et al. “Flipping of the Tidal Tails of the Ophiuchus Stream due to the Decelerating Galactic Bar”, 2025, ApJ, 984, 189. [\[link\]](#)
 44. Li, S., Wang, W., Koposov, S. E., **Li, T. S.**, et al. “SpecDis: Value Added Distance Catalog for 4 Million Stars from DESI Year-1 Data”, 2025, AJ, 170, 171. [\[link\]](#)
 43. Atzberger, K. R., Usman, S. A., Ji, A. P., Cullinane, L. R., Erkal, D., Hansen, T. T., Lewis, G. F., **Li, T. S.**, et al. “Chemical Abundances in the Leiptr Stellar Stream: A Disrupted Ultra-faint Dwarf Galaxy?”, 2025, OJAp, 8, 68. [\[link\]](#)

42. Byström, A., Koposov, S. E., Lilleengen, S., **Li, T. S.**, et al. “Exploring the interaction between the MW and LMC with a large sample of blue horizontal branch stars from the DESI survey”, 2025, MNRAS, 542, 560. [\[link\]](#)
41. Danieli, S., Kado-Fong, E., Huang, S., Luo, Y., **Li, T. S.**, et al. “First Data Release of the Merian Survey: A Wide-field Imaging Survey of Dwarf Galaxies at $z \sim 0.06\text{--}0.10$ ”, 2025, ApJ, 993, 110. [\[link\]](#)
40. Li, Z., Han, J., Wang, W., Qian, Y.-Z., Li, Q., Jing, Y., **Li, T. S.** “EMPDF: inferring the Milky Way mass with data-driven distribution function in phase space”, 2025, MNRAS, 538, 1442. [\[link\]](#)
39. Tan, C. Y., Cerny, W., Drlica-Wagner, A., Pace, A. B., Geha, M., Ji, A. P., **Li, T. S.**, et al. “A Pride of Satellites in the Constellation Leo? Discovery of the Leo VI Milky Way Satellite Ultra-faint Dwarf Galaxy with DELVE Early Data Release 3”, 2025, ApJ, 979, 176. [\[link\]](#)
38. Valluri, M., Fagrelus, P., Koposov, S. E., **Li, T. S.**, et al. “GD-1 Stellar Stream and Cocoon in the DESI Early Data Release”, 2025, ApJ, 980, 71. [\[link\]](#)
37. Pan, Y., Chiti, A., Drlica-Wagner, A., Ji, A. P., **Li, T. S.**, et al. “Stellar Metallicities from DECam u -band Photometry: A Study of Milky Way Ultra-faint Dwarf Galaxies”, 2025, ApJ, 978, 39. [\[link\]](#)
36. Koposov, S. E., Allende Prieto, C., Cooper, A. P., **Li, T. S.**, et al. “DESI Early Data Release Milky Way Survey value-added catalogue”, 2024, MNRAS, 533, 1012. [\[link\]](#)
35. Hilmi, T., Erkal, D., Koposov, S. E., **Li, T. S.**, et al. “Inferring dark matter subhalo properties from simulated subhalo-stream encounters”, 2024, submitted to MNRAS, [arXiv:2404.02953](#).
34. Luo, Y., Leauthaud, A., Greene, J., Huang, S., Kado-Fong, E., Danieli, S., **Li, T. S.**, et al. “The Merian Survey: Design, Construction, and Characterization of a Filter Set Optimized to Find Dwarf Galaxies and Measure their Dark Matter Halo Properties with Weak Lensing”, 2024, MNRAS, 530, 4988. [\[link\]](#)
33. Fan, T. J., Moon, D.-S., Park, H. S., et al. (including **Li, T. S.**) “Dwarf galaxy discoveries from the KMTNet supernova programme – III. The Milky-Way analogue NGC 2997 group”, 2023, MNRAS, 525, 4904. [\[link\]](#)
32. Cerny, W., Martínez-Vázquez, C. E., Drlica-Wagner, A., et al. (including **Li, T. S.**) “Six More Ultra-Faint Milky Way Companions Discovered in the DECam Local Volume Exploration Survey”, 2023, ApJ, 953, 1. [\[link\]](#)
31. Shipp, N., Panithanpaisal, N., Necib, L., et al. (including **Li, T. S.**) “Streams on FIRE: Populations of Detectable Stellar Streams in the Milky Way and FIRE”, 2023, ApJ, 949, 44. [\[link\]](#)
30. Simon, J. D., Brown, T. M., Mutlu-Pakdil, B., et al. (including **Li, T. S.**) “Timing the r-process Enrichment of the Ultra-faint Dwarf Galaxy Reticulum II”, 2023, ApJ, 944, 43. [\[link\]](#)
29. Lilleengen, S., Petersen, M. S., Erkal, D., et al. (including **Li, T. S.**) “The effect of the deforming dark matter haloes of the Milky Way and the Large Magellanic Cloud on the Orphan-Chenab stream”, 2023, MNRAS, 518, 774. [\[link\]](#)
28. Chiti, A., Simon, J. D., Frebel, A., et al. (including **Li, T. S.**) “Magellan/IMACS Spectroscopy of Grus I: A Low Metallicity Ultra-faint Dwarf Galaxy”, 2022, ApJ, 939, 41. [\[link\]](#)
27. Mau, S., Nadler, E. O., Wechsler, R. H., et al. (including **Li, T. S.**) “Milky Way Satellite Census. IV. Constraints on Decaying Dark Matter from Observations of Milky Way Satellite Galaxies”, 2022, ApJ, 932, 128. [\[link\]](#)
26. Yuan, Z., Malhan, K., Sestito, F., et al. (including **Li, T. S.**) “The Complexity of the Cetus Stream Unveiled from the Fusion of STREAMFINDER and StarGO”, 2022, ApJ, 930, 103. [\[link\]](#)
25. Vivas, A. K., Martinez-Vazquez, C. E., Walker, A. R., Belokurov, V., **Li, T. S.**, et al. “Variable Stars in the Giant Satellite Galaxy Antlia 2”, 2022, ApJ, 926, 78. [\[link\]](#)
24. Ferguson, P. S., Shipp, N., Drlica-Wagner, A., **Li, T. S.**, et al. “DELVE-ing into the Jet: A Thin Stellar Stream on a Retrograde Orbit at 30 kpc”, 2022, AJ, 163, 18. [\[link\]](#)
23. Shipp, N., Erkal, D., Drlica-Wagner, A., **Li, T. S.**, et al. “Measuring the Mass of the Large Magellanic Cloud with Stellar Streams Observed by S^5 ”, 2021, ApJ, 923, 149. [\[link\]](#)
22. Casey, A. R., Ji, A. P., Hansen, T. T., **Li, T. S.**, et al. “Signature of a Massive Rotating Metal-poor Star Imprinted in the Phoenix Stellar Stream”, 2021, ApJ, 921, 67. [\[link\]](#)
21. Hansen, T. T., Ji, A. P., Da Costa, G. S., **Li, T. S.**, et al. “ S^5 : The Destruction of a Bright Dwarf Galaxy as Revealed by the Chemistry of the Indus Stellar Stream”, 2021, ApJ, 915, 103. [\[link\]](#)
20. Lu, W., Fuller, J., Raveh, Y., Perets, H. B., **Li, T. S.**, et al. “The former companion of hyper-velocity star S5-HVS1”, 2021, MNRAS, 503, 603. [\[link\]](#)
19. Tanoglidis, D., Drlica-Wagner, A., Wei, K., **Li, T. S.**, et al. “Shadows in the Dark: Low-surface-brightness Galaxies Discovered in the Dark Energy Survey”, 2021, ApJS, 252, 18. [\[link\]](#)

18. Simon, J. D., Brown, T. M., Drlica-Wagner, **Li, T. S.**, et al. “Eridanus II: A Fossil from Reionization with an Off-center Star Cluster”, 2021, ApJ, 908, 18. [\[link\]](#)
17. Chang, J., Yuan, Z., Xue, X.-X., Simion, I. T., Kang, X., **Li, T. S.**, et al. “Is NGC 5824 the Core of the Progenitor of the Cetus Stream?”, 2020, ApJ, 905, 100. [\[link\]](#)
16. Hansen, T. T., Marshall, J. L., Simon, J. D., **Li, T. S.**, et al. “Chemical Analysis of the Ultrafaint Dwarf Galaxy Grus II. Signature of High-mass Stellar Nucleosynthesis”, 2020, ApJ, 897, 183. [\[link\]](#)
15. Li, Z.-Z., Qian, Y.-Z., Han, J., **Li, T. S.**, et al. “Constraining the Milky Way Mass Profile with Phase-space Distribution of Satellite Galaxies”, 2020, ApJ, 894, 10. [\[link\]](#)
14. Drlica-Wagner, A., Bechtol, K., Mau, S., McNanna, M., Nadler, E. O., Pace, A. B., **Li, T. S.**, et al. “Milky Way Satellite Census. I. The Observational Selection Function for Milky Way Satellites in DES Y3 and Pan-STARRS DR1”, 2020, ApJ, 893, 47. [\[link\]](#)
13. Allende Prieto, C., Cooper, A. P., Dey, A., Gansicke, B. T., Koposov, S. E., **Li, T. S.**, et al. “Preliminary Target Selection for the DESI Milky Way Survey (MWS)”, 2020, Research Notes of the AAS, 4, 10. [\[link\]](#)
12. Marshall, J. L., Hansen, T., Simon, J. D., **Li, T. S.**, et al. “Chemical Abundance Analysis of Tucana III, the Second r-process Enhanced Ultra-faint Dwarf Galaxy”, 2019, ApJ, 882, 177. [\[link\]](#)
11. Torrealba, G., Belokurov, V., Koposov, S. E., **Li, T. S.**, et al. “The hidden giant: discovery of an enormous Galactic dwarf satellite in Gaia DR2”, 2019, MNRAS, 488, 2743. [\[link\]](#)
10. Erkal, D., Belokurov, V., Laporte, C. F. P., Koposov, S. E., **Li, T. S.**, et al. “The total mass of the Large Magellanic Cloud from its perturbation on the Orphan stream”, 2019, MNRAS, 487, 2685. [\[link\]](#)
9. Mau, S., Drlica-Wagner, A., Bechtol, K., Pace, A. B., **Li, T. S.**, et al. “A Faint Halo Star Cluster Discovered in the Blanco Imaging of the Southern Sky Survey”, 2019, ApJ, 875, 154. [\[link\]](#)
8. Wang, M. Y., de Boer, T., Pieres, A., **Li, T. S.**, et al. “The Morphology and Structure of Stellar Populations in the Fornax Dwarf Spheroidal Galaxy from Dark Energy Survey Data”, 2019, ApJ, 881, 118. [\[link\]](#)
7. Wang, M. Y., Koposov, S., Drlica-Wagner, A., Pieres, A., **Li, T. S.** “Rediscovery of the Sixth Star Cluster in the Fornax Dwarf Spheroidal Galaxy”, 2019, ApJL, 875, L13. [\[link\]](#)
6. Shipp, N., Drlica-Wagner, A., Balbinot, E., Ferguson, P., Erkal, D., **Li, T. S.**, et al. “Stellar Streams Discovered in the Dark Energy Survey”, 2018, ApJ, 862, 114. [\[link\]](#)
5. Bernstein, G. M., Abbott, T. M. C., Armstrong, R., Burke, D. L., Diehl, H. T., Gruendl, R. A., Johnson, M. D., **Li, T. S.**, et al. “Photometric Characterization of the Dark Energy Camera”, 2018, PASP, 130, 4501. [\[link\]](#)
4. Bergemann, M., Sesar, B., Cohen, J., Serenelli, A. M., Sheffield, A. A., **Li, T. S.**, et al. “Two chemically similar stellar overdensities on opposite sides of the plane of the Galactic disk”, 2018, Nature, 555, 334. [\[link\]](#)
3. Johnston, K. V., Price-Whelan, A. M., Bergemann, M., Laporte, C., **Li, T. S.**, et al. “Disk Heating, Galactoseismology, and the Formation of Stellar Halos”, 2017, Galaxies, 5, 44. [\[link\]](#)
2. Hansen, T. T., Simon, J. D., Marshall, J. L., **Li, T. S.**, et al. “An r-process Enhanced Star in the Dwarf Galaxy Tucana III”, 2017, ApJ, 838, 44. [\[link\]](#)
1. Melendez, J., Placco, V. M., Tucci-Maia, M., Ramirez, I., **Li, T. S.** “2MASS J18082002-5104378: The Brightest (V=11.9) Ultra Metal-Poor Star”, 2016, A&A, 585, 5. [\[link\]](#)

Non-Refereed

Total: 29 non-refereed publications and reports (2012 – 2026): 13 conference proceedings and 16 white papers, mission/facility papers, reports, and book chapters.

Conference Proceedings

13. Pratsos, A., Dey, B., **Li, T. S.** “Characterizing Stellar Streams with Error-Aware Machine Learning”, 2026, Proc. of the Conference on Physics and AI at Stanford University (PAI2026), [arXiv:2606.09576](#).
12. Xu, J. E., Jeram, S., Chen, S., **Li, T. S.**, et al. “Prototyping the Next-Generation Spectrograph for Space Exploration: Digital Micromirror Device-based Multi-Object Spectrograph (DMD-MOS) Calibration and Performance Assessment”, 2026, Proc. of SPIE (Astronomical Telescopes + Instrumentation, Copenhagen), in press.
11. Chen, S., Jeram, S., Xu, J. E., Yao, X., Barnett, M., **Li, T. S.**, et al. “Prototyping the Next-Generation Spectrograph for Space Exploration: DMD-Based Multi-Object Spectrograph Design and Production”, 2026, Proc. of SPIE, in press.
10. Singh, D., **Li, T. S.**, Jeram, S., Chen, S. “Automated multi-stressor optical fiber focal ratio degradation

and throughput characterization system for future massive multiplex spectroscopic survey”, 2026, Proc. of SPIE, in press.

9. Azam, Z., Khandelwal, A., Dungee, R., Li, T. S., Jeram, S., Chen, S. “Measuring the linear gain performance for the OrcaQuest CMOS imaging sensor”, 2026, Proc. of SPIE, in press.
8. Dungee, R., Sivanandam, S., Chun, M. R., et al. (including Li, T.) “Ae’o: a SCIDAR instrument for Maunakea”, 2026, Proc. of SPIE, in press.
7. Khandelwal, A., et al. (including Li, T. S.) “Beyond CCDs: Characterization of sCMOS detectors for optical astronomy”, 2024, Proc. of SPIE Vol. 13103, 131030R. [\[link\]](#)
6. Meisner, A. M., et al. (including Li, T. S.) “Performance of Kitt Peak’s Mayall 4-meter telescope during DESI commissioning”, 2020, Proc. of SPIE Vol. 11447, 1144794. [\[link\]](#)
5. Drlica-Wagner, A., Marrufo Villalpando, E., O’Neil, J., Estrada, J., Holland, S., Kurinsky, N., Li, T. S., “Characterization of skipper CCDs for cosmological applications”, 2020, Proc. of SPIE Vol. 11454, 114541A. [\[link\]](#)
4. Li, T. S., DePoy, D. L., Marshall, J. L., Nagasawa, D. Q., Carona, D. W., Boada, S., “Monitoring the atmospheric throughput at Cerro Tololo Inter-American Observatory with aTmCam”, 2014, Proc. of SPIE Vol. 9147, 91476Z. [\[link\]](#)
3. DePoy, D. L., Allen, R., Li, T. S., Marshall, J. L., Papovich, C., Prochaska, T., Shectman, S., “An update on the wide field, multi-object, moderate-resolution, spectrograph for the Giant Magellan Telescope”, 2014, Proc. of SPIE Vol. 9147, 914720. [\[link\]](#)
2. Marshall, J. L., DePoy, D. L., Prochaska, T., Allen, R. D., Williams, P., Rheault, J.-P., Li, T. S., and 22 colleagues, “VIRUS instrument collimator assembly”, 2014, Proc. of SPIE Vol. 9147, 91473S. [\[link\]](#)
1. Li, T. S., DePoy, D. L., Kessler, R., Burke, D. L., Marshall, J. L., Wise, J., Rheault, J.-P., Carona, D. W., Boada, S., Prochaska, T., Allen, R., “aTmcam: a simple atmospheric transmission monitoring camera for sub 1% photometric precision”, 2012, Proc. of SPIE Vol. 8446, 84462L. [\[link\]](#)

White Papers, Reports, and Book Chapters

9. Co-author, *The CASTOR Mission* (Côté et al.), 2025, JATIS, 11, 042202. [\[link\]](#)
8. Co-author, *The Spectroscopic Stage-5 Experiment* (Spec-S5), 2025, [arXiv:2503.07923](#).
7. Facilitator for Snowmass CF03 white papers: Chakrabarti, Drlica-Wagner, Li et al., *Snowmass2021 Cosmic Frontier White Paper: Observational Facilities to Study Dark Matter* ([arXiv:2203.06200](#)); Drlica-Wagner et al., *Report of the Topical Group on Cosmic Probes of Dark Matter for Snowmass 2021* ([arXiv:2209.08215](#)).
6. Author for 7 Astro2020 Science White Papers, including one as lead author: Li et al., *Dark Matter Physics with Wide Field Spectroscopic Surveys* (**BAAS**, 51, 252); Dey et al., *Mass Spectroscopy of the Milky Way* (**BAAS**, 51, 489); Simon et al., *Dynamical Masses for a Complete Census of Local Dwarf Galaxies* ([arXiv:1903.04743](#)); Simon et al., *Testing the Nature of Dark Matter with Extremely Large Telescopes* ([arXiv:1903.04742](#)); Sanderson et al., *The Multidimensional Milky Way* ([arXiv:1903.07641](#)); Ji et al., *Local Dwarf Galaxy Archaeology* ([arXiv:1903.09275](#)); Bechtol et al., *Dark Matter Science in the Era of LSST* ([arXiv:1903.04425](#)).
5. Chapter Leader, *The Detailed Science Case for the Maunakea Spectroscopic Explorer*, 2019 edition ([arXiv:1904.04907](#); [arXiv:1903.03155](#) for the chapter).
4. Chapter Author, *Probing the Fundamental Nature of Dark Matter with the Large Synoptic Survey Telescope*, 2019, [arXiv:1902.01055](#).
3. Organizer and Chapter Author, *Petabytes to Science*, 2019, [arXiv:1905.05116](#).
2. Chapter Author, *GMT Science Book 2018*. [\[link\]](#)
1. Chapter Author, *Maximizing Science in the Era of LSST: A Community-Based Study of Needed US Capabilities*, 2016, [arXiv:1610.01661](#).

Presentations

Keynote and Invited Plenary/Review Talks (selected)

- Invited Keynote Talk, IAU Symposium 403 “The Hidden Beauty of the Galactic Outskirts”, Cordoba, Spain, Oct 2025.
- Invited Review Talk, IAU Symposium 395 “Stellar populations in the Milky Way and beyond”, Paraty, Brazil, Nov 2024.

- Invited Review Talk, KICP Workshop “Dwarf Galaxies, Star Clusters, and Streams in the LSST Era”, University of Chicago, IL, Jul 2024.
- Invited Plenary Talk, DESI Collaboration Meeting, Durham University, Durham, UK, Jul 2023.
- Invited Review Talk, Pollica Workshop “Self-Interacting Dark Matter: Models, Simulations and Signals”, Pollica, Italy, Jun 2023.
- Invited Topical Talk, IAU Symposium 379 “Dynamical Masses of Local Group Galaxies”, Potsdam, Germany, Mar 2023.
- Invited Plenary Talk, TeV Particle Astrophysics 2021 Conference, Chengdu, China, Oct 2021.
- Invited Plenary Talk, SnowPAC “Big Questions, Big Surveys, Big Data: Astronomy & Cosmology in the 2020s”, Snowbird, UT, May 2018.

Invited Lectures, Colloquia, and Seminars

- Colloquium, Australian National University, Australia, Feb 2026.
- Colloquium, Universidad de Chile, Santiago, Chile, Oct 2025.
- Colloquium, Cerro Tololo Inter-American Observatory, La Serena, Chile, Sept 2025.
- Lecture, “EXPLORE Summer School and Conference 2025”, York University, Toronto, Canada, Aug 2025 (1 hr lecture + 2 hr tutorials).
- Colloquium, Lund University, Sweden, Jun 2025.
- Astroseminar, City University of Hong Kong, China, May 2025.
- Invited Talk, Workshop “Cosmological Probes of New Physics”, Notre Dame, USA, Apr 2025.
- Colloquium, Department of Physics & Astronomy, McMaster University, Canada, Feb 2025.
- Colloquium, Department of Astronomy, University of Arizona, Tucson, AZ, Nov 2024.
- Invited Talk, Conference “Galphases24: Accretion in the Local Group”, Strasbourg, France, Aug 2024.
- Colloquium, Astronomisches Rechen-Institut, Heidelberg University, Germany, Aug 2024.
- Invited long-term participant, KITP Program “Dark Matter Theory, Simulation, and Analysis in the Era of Large Surveys”, UC Santa Barbara, CA, Jun 2024.
- Invited Talk, KITP Conference “Cosmic Signals of Dark Matter Physics: New Synergies”, UC Santa Barbara, CA, Jun 2024.
- Invited Astroseminar Talk, Waterloo Centre for Astrophysics, University of Waterloo, ON, Mar 2024.
- Colloquium, Department of Astronomy, Ohio State University, Columbus, OH, Jan 2024.
- Colloquium, Department of Astronomy and Physics, Saint Mary’s University, Halifax, NS, Nov 2023.
- Colloquium, Department of Physics, Engineering Physics & Astronomy, Queen’s University, Kingston, ON, Oct 2023.
- Colloquium, Department of Astronomy, UC Berkeley, Berkeley, CA, Sept 2023.
- Colloquium, IAS/Department of Astrophysical Sciences, Princeton University, Princeton, NJ, Sept 2023.
- Invited Seminar Talk, Durham University, UK, Jul 2023.
- Invited Seminar Talk, University of Surrey, UK, Jul 2023.
- Colloquium, Kapteyn Astronomical Institute, University of Groningen, the Netherlands, Jul 2023.
- Colloquium, Leiden Observatory, Leiden University, the Netherlands, Jul 2023.
- Colloquium, National Astronomical Observatories of China (NAOC), Beijing, China, Jun 2023.
- Colloquium, Kavli Institute for Astronomy and Astrophysics (KIAA), Peking University, Beijing, China, Jun 2023.
- Invited Virtual Seminar, AAS Division on Dynamical Astronomy Community Seminar ([link to talk](#)), Mar 2023.
- Invited Seminar, University of Victoria, Victoria, BC, Feb 2023.
- Invited Seminar, National Research Council of Canada’s Herzberg Astronomy & Astrophysics Research Centre, Victoria, BC, Feb 2023.
- Colloquium, Department of Physics, University of Michigan, Ann Arbor, MI, Nov 2022.
- Astrophysics Colloquium, MIT Kavli Institute, Cambridge, MA, Nov 2022.
- Colloquium, Department of Physics, University of Toronto, Toronto, Canada, Sept 2022.
- Invited Seminar Talk, Texas A&M University, College Station, TX, Aug 2022.
- Invited Seminar Talk, University of Notre Dame, Notre Dame, IN, Apr 2022.
- (remote) Colloquium, Indiana University, Bloomington, IN, Oct 2021.
- (remote) Colloquium, NRC Herzberg Astronomy & Astrophysics Research Centre, Victoria, BC, Sept 2021.
- (remote) Invited Seminar Talk, N3AS ([link to talk](#)), May 2021.

- (remote) Colloquium, University of British Columbia, Vancouver, Canada, Apr 2021.
- (remote) Invited Seminar Talk, University of Kentucky, Lexington, KY, Feb 2021.
- (remote) Invited Seminar Talk, UC Davis, Davis, CA, Jan 2021.
- (remote) Invited Seminar Talk, University College London, England, Nov 2020.
- (remote) Invited Seminar Talk, McGill Space Institute, Montreal, Canada, Oct 2020.
- Colloquium, UC Santa Cruz, Santa Cruz, CA, Nov 2019.
- Colloquium, University of Toronto, Toronto, Canada, Oct 2019.
- Invited Talk, LSST Dark Matter Workshop, University of Chicago, Chicago, IL, Aug 2019.
- Invited Talk, IAU Symposium 353 “Galactic Dynamics in the Era of Large Surveys”, Shanghai, China, Jul 2019.
- Invited Talk, Conference “Science in our own Backyard: Exploring the Galaxy and the Local Group with WFIRST”, Caltech, Pasadena, CA, Jun 2019.
- Invited Talk, KITP Workshop “In the Balance: Stasis and Disequilibrium in the Milky Way”, Santa Barbara, CA, Apr 2019.

Conference and Symposia Contributed Talks / Posters

- Contributed Talk (and SOC member), Workshop on “Small-Scale Structure of the Universe and Self-Interacting Dark Matter”, Valencia, Spain, Jun 2025.
- Contributed Talk, Workshop “Great Lakes Clusters and Streams”, University of Michigan, Ann Arbor, MI, Aug 2023.
- Contributed Talk, Conference “MODEST-23: Star Clusters in the Post-Pandemic Era”, Northwestern University, Evanston, IL, Aug 2023.
- Contributed Talk, DECam at 10 Years Workshop, Tucson, AZ, Sept 2022.
- [iPoster](#), 235th AAS Meeting, Honolulu, Hawaii, Jan 2020.
- Contributed Talk, Conference “Massively Multiplexed Spectroscopy with MSE: Science, Project and Vision”, Tucson, AZ, Feb 2019.
- Contributed Talk, Conference “Shedding Light on the Dark Universe with Extremely Large Telescopes”, Lanzhou, China, Aug 2017.

Media and Other Public Engagements

Press Coverage

- **May 2026 – CFI/ORF Funding for ANDES: “Canada Now at the Heart of the World’s Largest Telescope with the ANDES Instrument.”** [Dunlap Institute press release](#); joint announcement with Université de Montréal and UBC.
- **Mar 2025 – DESI Dark Energy Results: “Largest 3D Map of the Universe Points to Evolving Dark Energy.”** [U of T Arts & Science / FAS press release](#); featured in *TIME* magazine, *The Guardian*, NOIRLab, Berkeley Lab News Center, and more.
- **Jan 2022 – One Dozen Stellar Streams: “Twelve for Dinner: The Milky Way’s Feeding Habits Shine a Light on Dark Matter.”** [Dunlap press release](#); [U of T News](#); featured in *APS Physics*, *The Globe and Mail*, SYFY Wire, Sky & Telescope, Inverse, C|net, and more. [Press conference at AAS 239](#). Images/videos: [S5 release page](#).
- **Jul 2020 – Phoenix Stream: “The Ancient Stars That Time Forgot.”** [Macquarie University / University of Sydney press release](#); featured in *Nature*, ABC Radio National, C|net, Space Australia, The Register, and Vice. Images/videos: [S5 release page](#).
- **Nov 2019 – S5-HVS1: “Discovery of a 1700 km/s Star from the Galactic Center.”** [Carnegie Science press release](#); covered by *The New York Times*, CBS News, *The Independent*, *Forbes*, *The Guardian*, *Newsweek*, and ABC. Images/videos: [S5 release page](#).

Public Talks and Outreach

- Speaker, Astronomy on Tap (2018 – present) — public astronomy talks in bars and pubs across multiple cities, with audiences of 30–400 per event.
- Starfest 2025 Keynote Speaker (North York Astronomical Association). “Illuminating the Dark Universe: How Cosmic Surveys Uncover Dark Matter (and Black Holes)”, Aug 23, 2025 (500+ audience).
- See Talk (via Speaker Bureau). “From Ancient to Modern Cosmology with Astronomical Observations”, Aug

7, 2025.

- Toronto Public Library Talk (via Speaker Bureau). “From Ancient to Modern Cosmology with Astronomical Observations”, Feb 27, 2026.
- Presenter, Star Talks, Astronomy & Space Exploration Association, University of Toronto, 2024.
- Instructor, Vision of Science Summer Intern Program, 2022.
- Lecturer, Fermilab Lifelong Learning Institute (LLI) Program, 2018 – 2019.
- Lecturer, Fermilab Saturday Morning Physics for High School Students, 2017 – 2019.
- Coordinator, Fermilab Saturday Morning Physics for High School Students, 2016 – 2019.
- E/PO representative, Dark Energy Survey, 2014 – 2016.
- Coordinator, Science Olympiad, 2014.
- Instructor, “Expanding Your Horizons” Workshop, 2012.
- Organizer, Texas A&M Physics and Astronomy Festival, 2011 – 2014.
- Organizer, Texas A&M Star Parties, 2011 – 2013.

Teaching Experience

Undergraduate

University of Toronto:

AST121H – The Origin and Evolution of the Universe (Winter 2022).

- This introductory course is aimed at students enrolling in science and engineering majors, but no prior knowledge of astronomy is assumed. The purpose of the course is to provide students with an overall cosmic perspective on the origin and evolution of the Universe and all that it contains — from the chemical elements, to stars and galaxies, and life. It is intended to share how exciting modern astronomy and astrophysics is and to help students understand the big questions astronomers are probing.

AST301H – Observational Astronomy (Fall 2021, 2022, 2023, 2024).

- In this course, students use hands-on activities to explore the methods modern astronomers use to answer some of the Big Questions. Using some of the same equipment used by professional astronomers, students make their own observations of the cosmos. They study stars as they explode in real time, reveal the motions of galaxies, and prove to themselves that the universe is getting bigger every day. The course is intended for students with majors outside the physical sciences who want to gain experience with hands-on observational astronomy. Basic arithmetic is used, but no advanced mathematics. Most of the math the students need is done via simple programs in the Python programming language, for which templates are provided. No previous experience with programming is required.

ROP 299/399H – Research Opportunity Program (Summer 2024, Summer 2025). Undergraduate research supervision through the Department ROP program. I take 3–4 students each term and have an independent project for each student with tailored onboarding and training plans.

Scheduled for 2026–27: AST326Y1 – Practical Astronomy (Fall 2026 – Winter 2027, instructor): a hands-on laboratory course in modern astronomical measurement – detectors and noise, photometry and astrometry, spectroscopy, and observations with the campus telescopes – along with the statistics, error analysis, signal processing, and scientific writing at the foundation of experimental astrophysics.

Graduate

University of Toronto:

- **AST1420H – Galactic Structure and Dynamics** (Winter 2026). Graduate-level course covering gravitational potentials and classical mechanics, orbits in spherical and axisymmetric potentials, collisionless stellar systems and the Jeans equations, galactic disks, dynamical friction and mergers, chemical evolution, stellar population synthesis, and the formation and evolution of dark matter halos. As a major revision, I developed two entirely new student projects (“Jeans Modeling of Milky Way Satellite Galaxies” and “Chemical Evolution of Milky Way Satellite Galaxies”), each accompanied by introductory material and a tutorial Jupyter notebook, enabling students to apply classroom concepts in a realistic research setting.
- **AST3101H – Practical Skills for Astronomical Observation: from Proposal Writing to Observations** (Winter 2024; mini course; *designed and created by me*). Mini-course that I designed and created

for first- and second-year graduate students, covering the full end-to-end observing workflow: formulating a science question, writing and defending an observing proposal, understanding how TAC review panels work, planning an observing run, and conducting observations. Students write their own observing proposal and form mock anonymous review panels to referee one another’s submissions. Students also participate in real remote observing runs on Blanco, Magellan, and Keck to experience how observations are conducted in practice.

Supervisory Experience

Postdoctoral Fellows

2026 – present	Tri Nguyen — NSERC Postdoctoral Fellow / Dunlap Fellow (starting Fall 2026)
2025 – present	Biprateep Dey — Schmidt AI / CITA / Dunlap Fellow
2025 – present	Kevin McKinnon — Schmidt AI Fellow
2024 – present	Ryan Dungee — Dunlap Instrumentation Fellow
2023 – present	Gustavo Medina — personal postdoc (Jan–Apr 2023), A&S Fellow (May 2023 – Apr 2025), Dunlap Fellow (May 2025 – present); <i>starting a new postdoc as a Veni Fellow at Kapteyn Astronomical Institute in Groningen in Fall 2026.</i>
2023 – present	Nathan Sandford — joint personal postdoc with Jo Bovy (Sept 2023 - Aug 2024), A&S Fellow (Sept 2024 – Aug 2026); <i>starting a new postdoc as a US NSF Fellow at UC Santa Cruz in Fall 2026.</i>

Graduate Student Supervision

Primary/Co-Advisor: PhD Students

2025 – present	Anika Slizewski — co-advisor (co-supervise with Gwen Eadie). <i>Thesis:</i> Mass Estimators with phase-space distribution function.
2024 – present	Nasser Mohammed — primary advisor. <i>Thesis:</i> Constrain Dark Matter with Stellar Streams in DESI.
2022 – 2026	Mairead Heiger — primary advisor (co-supervised with Josh Speagle). <i>Thesis:</i> Galactic Chemical Evolution with Dwarf Galaxies and Metal-poor Stars; defended August 2026; <i>starting a new postdoc as a GECO/CosmicAI Fellow at the University of Virginia in Fall 2026.</i>
2023 – 2025	Petra Awad (University of Groningen) — co-advised one paper of her PhD thesis: the S^5 deep spectroscopic study of globular cluster tidal tails; <i>now a postdoc at Leiden Observatory since 2025.</i>
2017 – 2021	Nora Shipp (University of Chicago) — PhD co-advisor; research on stellar streams in the Dark Energy Survey and S^5 ; <i>now faculty at the University of Washington since 2024.</i>

Primary/Co-Advisor: Master’s (or equivalent) Students

2025 – present	Emma Jiayi Xu (Physics Department) — co-advisor (co-supervise with Barth Netterfield). <i>Project:</i> Optimize the filter selections for GigaBIT for stellar parameters and abundance determination.
2024 – 2026	Emma Jarvis — primary supervisor for AST 1500 project. <i>Project:</i> Characterizing the GD-1 Stream with DESI DR2 Data: Thin Stream and Hot Cocoon.
2024 – 2025	Ethen Sun — primary supervisor for AST 1500 project. <i>Project:</i> On-sky testing of sCMOS cameras at the E.C. Carr Astronomical Observatory.

Graduate Student Supervisory Committees (PhD, University of Toronto)

- Committee Member, **Grace Connolly** (2026–present).
- Committee Member, **Qing Liu** (2021–2024)
- Non-Supervisory Committee Member, **Steffani Grondin**, Final Oral Examination (2025).
- Non-Supervisory Committee Member, **Nathaniel Starkman**, Final Oral Examination (2024).
- Non-Supervisory Committee Member, **Aarya Patil**, Final Oral Examination (2023).
- Thesis Qualifying Exam Committee Member, **Abigail Denney** (2026).
- Thesis Qualifying Exam Committee Member, **Anika Slizewski** (2024).

- Thesis Qualifying Exam Committee Member, **Samantha Berek** (2022).
- Thesis Qualifying Exam Committee Member, **Steffani Grondin** (2022).
- Thesis Qualifying Exam Committee Member, **Jacob Taylor** (2021).

Undergraduate Student Research Supervision

2025 – 2026 Cohort (DAA, U of Toronto)

- **Laly Boyer** — Mitacs Globalink 2026: Fiber-based multi-object spectrograph (May 2026 – present).
- **Emily Moran** — Mitacs Globalink 2026: Stellar stream modeling with mock data (May 2026 – present).
- **Ouyun Shen** — Mitacs Globalink 2026: Dwarf galaxies and self-interacting dark matter (May 2026 – present).
- **Nawaf Alrefaie** — SUDS/KAUST summer intern 2026: Mapping dark matter with AI (May 2026 – present).
- **Yazan Bakhshwin** — SUDS/KAUST summer intern 2026: Metallicities of RR Lyrae stars with LSST (May 2026 – present).
- **Frank Ziang Chen** — SURP 2026: DMD-based multi-object spectrograph (May 2026 – present).
- **Matteo Moretti** — SURP 2026: The formation history of Andromeda (May 2026 – present).
- **Andy Yu** — SUDS 2026: Mapping dark matter with AI (May 2026 – present).
- **Damon Johnston** — SURP 2026: Stellar proper motions with HST (May 2026 – present).
- **Rosayla Coulthard** — ROP 2025 + AST 430: Carbon-Enhanced Metal-Poor Star Properties within Milky Way Halo Satellites (May 2025 – April 2026).
- **Alexandros Pratsos** — SUDS 2025 + AST 430: Identifying Milky Way Streams with Machine Learning (May 2025 – April 2026).
- **Nava Wolfish** — SURP 2025 + AST 425: Unraveling the Formation History of the Andromeda Galaxy with JWST (May 2025 – April 2026).
- **Zain Azam** — Work Study Program: CMOS detector characterization (May 2025 – April 2026).
- **Daksh Singh** — Work Study Program: Optical fiber focal ratio degradation and throughput characterization system for future massive multiplex spectroscopic survey (May 2025 – April 2026).
- **Liwen Chen** — ROP 2025 + SURP 2026: Pushing Stellar Motion Limits with James Webb Space Telescope and Gaia (May 2025 – present).
- **Mingzhi Jiang** — ROP 2025: Search for RR Lyrae stars in the Orphan-Chenab Stream with DECam (May 2025 – Apr 2026).
- **Wesley Luo** — SURP 2025 (with USRA) + AST 430: High-precision stellar motions and blue horizontal branch stars in the Local Universe (May 2025 – present).
- **Aishani Chaudhuri** — SURP 2025 (with UTEA): Hunting for Binary Stars in DESI Multi-epoch Spectroscopy (May 2025 – Apr 2026).
- **Emma Jiayi Xu** — SURP 2025: Prototyping the Next-Generation Spectrograph for Space Exploration (May – Aug 2025).
- **Juan Carlos Aranda Muñoz** — Mitacs Globalink 2025: Forecasting Dwarf Galaxy Dark Matter Profile Constraints for DESI (May – Aug 2025).
- **Alejandra Calderón Linares** — Mitacs Globalink 2025: Searching for Stellar Streams with 10 Million Stars in the Milky Way (May – Aug 2025).
- **Isabelle Laing** — SURP 2024 + AST 430 (co-supervised with Josh Speagle): Stellar chemical abundances with machine learning (May 2024 – Aug 2025).
- **Dhruv Pandya** — Work Study Program: Digital Micromirror Device-based Multi-Object Spectrograph (DMD-MOS) (Sept 2025 – April 2026).
- **Diego Martinez Noriega** — Waterloo Co-op (co-supervised with Josh Speagle): A Fast, Reproducible Framework to Yield Chemical Compositions from Spectroscopic Surveys (Sept 2025 – April 2026).

Former Undergraduate Mentees (selected)

- **Maia Wertheim** — PhD student at the University of Toronto from Fall 2026.
- **Joseph Tang** — PhD student at Columbia University since 2025; co-author on *Mohammed, Tang, Li et al. (2026)*.
- **Aditya Khandelwal** — PhD student at the University of Arizona since 2025; led *Khandelwal et al. (2024, SPIE)* and co-author on three 2026 SPIE papers.
- **Andrew Li** — Master’s student at the University of Victoria since 2025.
- **Peter Ma** — PhD student at UC Berkeley since 2024; led *Ma, Rogers, Li et al. (2025, ApJ)*.

- **Grace Yu** — PhD student at Caltech since 2024; led *Yu, Li et al. (2024, ApJ)*.
- **Jordan Bruce** — PhD student at Indiana University since 2023; led *Bruce, Li et al. (2023, ApJ)*.
- **William Cerny** — PhD student at Yale University since 2022; led *Cerny, Li et al. (2026, submitted)*, *Cerny, Drlica-Wagner, Li et al. (2023, ApJ)*, and *Cerny, Simon, Li et al. (2023, ApJ)*.
- **Sydney Jenkins** — PhD student at MIT since 2022; led *Jenkins, Li et al. (2021, ApJ)*.

Professional Service

Scientific Collaboration Leadership

- **Working Group Chair, DESI Milky Way Working Group** (2022 – 2025) — WG of ~50 active members; Builder since 2023, for contributions to the DESI focus-and-alignment system, commissioning, and science verification; Member since 2016.
- **Leader/Founder, Southern Stellar Stream Spectroscopic Survey (S^5)** (2018 – present) — ~40 active members.
- **Working Group Chair, Maunakea Spectroscopic Explorer (MSE) Dark Matter Working Group** (2018 – 2022).
- **Working Group Chair, DES Milky Way Working Group** (2018 – 2021) — WG of ~40 active members; Builder since 2015, for contributions to the DES calibration systems, commissioning, operation, and data releases; Member since 2012.
- **Member, Euclid Consortium** (2023 – present).
- **Member, SDSS-V** (2021 – present).
- **Builder, DECam Local Volume Exploration (DELVE)** (2025 – present); Member since 2018.
- **Member, CASTOR Science Team** (2022 – present).
- **Member, TMT/WFOS Science Team** (2021 – present).
- **Member, LSST Dark Energy Science Collaboration (DESC)** (2016 – present).

Advisory Boards, Committees, and Review Panels

- **SOC and LOC chair, “Near Field Cosmology in the Era of Big Data: Local Group and Beyond”**, Fields Institute, Toronto, Jul 2026 — 164 participants from 18 countries.
- **Chair, CFHT Community Survey Working Group** (2025 – present) — leading the 6-year community survey call and review process.
- **SOC, Spec-S5 Dark Matter Meeting**, Chicago, USA, Nov 2025.
- **SOC, Valencia Workshop on “the Small-Scale Structure of the Universe and Self-Interacting Dark Matter”**, Valencia, Spain, Jun 2025.
- **SOC, CFHT Users’ Meeting**, Montreal, Canada, May 2025.
- **SOC, CASCA Annual Conference**, Toronto, Jun 2024.
- **SOC, Workshop “Globular Clusters and Their Tidal Tails: From the Milky Way to the Local Group”**, Toronto, Jun 2024.
- **Canadian Representative, CFHT Science Advisory Committee (SAC)** (2023 – 2027) — includes two SAC meetings per year (Hawaii) and travel.
- **CADC (Canadian Astronomy Data Centre) User Board** (2023 – 2025).
- **Canadian Time Allocation Committee (TAC)** for Gemini and CFHT (2022 – 2024).
- **Roman Space Telescope 2022 Peer Review Panelist** (2023).
- **HST Cycle 30 Peer Review Panelist** (2022).
- **DESI Publication Board** (2021 – 2022).
- **Referee/Reviewer** for *Nature*, *ApJ*, *ApJL*, *MNRAS*, *A&A*, *PASP* (2015 – present).
- **Group Leader, DES Chromatic Correction and Interstellar Reddening Task Force** (2017 – 2019).
- **DES Early Career Scientist Committee** (2016 – 2018).

University Service

Department of Astronomy and Astrophysics (DAA) / Dunlap

2026 – present	Chair, Colloquium Committee (Dunlap/DAA)
2021 – present	Instrumentation Advisory Committee, Dunlap
2022 – 2026	Chair, TASTY Committee (Dunlap/DAA)
2025 – 2026	Graduate Qualifiers Committee, DAA
2025 – present	Training and Mentoring Committee, DAA
2021 – 2023, 2024 – 2025	Graduate Admissions Committee, DAA
2022 – 2023, 2025 – 2026	Faculty Search Committee, DAA
2022 – 2023	Dunlap Fellowship Committee, Dunlap
2021 – 2022	Colloquium Committee, DAA

Faculty of Arts & Science

2024 – present	Arts & Science Research Awards Adjudication Committee
2024 – present	New Researcher, Mid-Career Awards and McLean Awards Committee